

STATEMENT OF WORK

FOR

Cross-Connection Survey of Installation Water Systems

FAIRCHILD AIR FORCE BASE, WA

~~March 2026~~ **April 2026**

- 1 **PURPOSE.** This Statement of Work (SOW) outlines the cross-connection survey to conduct a comprehensive evaluation of Fairchild AFB's (FAFB) water system to protect potable water from potential contamination. The survey aims to ensure compliance with federal, state, and local regulations and industry best practices. The primary objectives of this SOW are to:
 - 1.1 Identify actual or potential cross-connections.
 - 1.2 Evaluate adequacy of protection of existing backflow prevention devices (BFDs).
 - 1.3 Identify deficiencies related to cross-connections and associated BFDs.
 - 1.4 Provide corrective action recommendations to protect the potable water system.
- 2 **APPLICABLE CODES AND STANDARDS.** The survey shall, at a minimum, comply with the following requirements, standards, and codes:
 - 2.1 Department of the Air Force Manual (DAFMAN) 32-1067, Water and Fuel Systems: Governs operation, maintenance, and protection of Air Force potable water systems.
 - 2.2 Air Force Instruction (AFI) 48-144, Drinking Water Surveillance Program: Ensures potable water safety and quality.
 - 2.3 Unified Facilities Criteria (UFC) 3-230-01, Water Storage, Distribution, and Transmission
 - 2.4 UFC 3-420-01, Plumbing Systems
 - 2.5 Washington Administrative Code (WAC) Title 246-290-490, Cross-Connection Control Requirements: Provides state-specific cross-connection control guidance.
 - 2.6 Uniform Plumbing Code (UPC): Industry standards for plumbing systems, including cross-connection control.
 - 2.7 American Water Works Association (AWWA) M14 Manual, Recommended Practice for Backflow Prevention and Cross-Connection Control: Sets industry standards for surveying and protecting water systems.
 - 2.8 University of Southern California (USC) Manual of Cross-Connection Control: Recognized reference for backflow prevention device standards.
 - 2.9 Washington State Department of Health (DoH) Guidance: Ensures compliance with state regulations for water system safety.
- 3 **SCOPE OF WORK.** The contractor shall, via physical observation and industry standard practices, conduct a comprehensive survey of the entirety of the premise (transmission and distribution lines) and in-premise protection, for each of Fairchild AFB's water systems, including the water systems at geographically separated units (GSUs). Include all vaults, Air Relief Valves (ARVs), insertion-extraction points, hose bibs (including RV sites, parks, etc), irrigation connections, etc. along the lines/system. Contractor will not be expected to dig to locate devices; FAFB will provide maps with known GIS coordinates of BFDs if available (reference Appendices B1, B2, and D).
 - 3.1 System Details: The following are not to be assumed as 100% accurate but rather only for approximation purposes for this SOW. The water system includes roughly:
 - 3.1.1 500 known backflow preventers (reference Appendix C1),

- 3.1.2 300+ buildings associated with Backflow Devices or potentially connected to the water systems in general, including GSUs (totaling 4.4M sqft, averaging 14.4k sqft per building) (reference Appendices C1 and C2).
- 3.1.3 390k+ linear feet of Main Line, which includes 60k+ linear feet of Main Transmission Line. (Appendix B1)
- 3.1.4 100k+ linear feet of Service Line (these numbers include both active and abandoned line lengths). (Appendix B1)
- 3.1.5 2 Medical and Dental clinic facilities (with 4 BFDs per the ASR) (App. B1).
- 3.2 The survey shall address the following areas:
 - 3.2.1 Premise and In-Premise Protection
 - 3.2.1.1 Inspect external and internal water system components to identify cross-connections.
 - 3.2.1.2 Evaluate adequacy of backflow prevention devices in use for compliance with applicable codes. (Physical evaluation of devices' function is not necessary.)
 - 3.2.2 Irrigation Systems
 - 3.2.2.1 Inspect irrigation systems for cross-connections.
 - 3.2.2.2 Evaluate the adequacy of backflow prevention measures for irrigation systems. (Physical evaluation of devices' function is not necessary.)
 - 3.2.2.3 Note 1: Some facilities may have irrigation BFDs within the building.
 - 3.2.2.4 Note 2: FAFB assumes there are upward of 10% additional connections unaccounted for on the irrigation system.
 - 3.2.3 Fire Protection Systems
 - 3.2.3.1 Inspect fire protection systems for potential cross-connections.
 - 3.2.3.2 Assess backflow prevention devices associated with these systems for proper selection and function. (Physical evaluation of devices' function is not necessary.)
 - 3.2.3.3 For non-potable fire suppression systems, stop evaluation at the cross-connection device.
 - 3.2.4 Geographically Separated Units (GSUs) encompass 5 separate water systems (Ft. Wright Wells & Cemetery and Geiger Reservoir make up 1 water system):
 - 3.2.4.1 Ft. Wright Wells and Cemetery (FAFB water system)
 - 3.2.4.2 Geiger Reservoir (FAFB water system with backup intertie to City of Spokane)
 - 3.2.4.3 White Bluff (independent water system)
 - 3.2.4.4 Ruby Creek (independent water system)
 - 3.2.4.5 Tacoma Creek (independent water system)
 - 3.2.4.6 Clear Lake Recreation Area (independent water system)
- 3.3 The survey shall not address Balfour Beatty-contracted housing units; survey stops at the last BFD before the distribution line solely supplies the housing communities.

4 SURVEY REQUIREMENTS. The contractor shall:

- 4.1 Conduct the survey between 01 July 2026 and 30 September 2026.

- 4.2 Physical observation and assessment of status of device protection (i.e. ARVs) along the entire route of the Main Transmission Line from the source wells at Ft. Wright to Geiger Reservoir to FAFB is to be performed between 01 Jul 2026 and 30 Sep 2026 to facilitate easier observation of potential leak or cross-connection sources. To reiterate, the entire route of this Main Transmission Line is to be physically observed for potential cross-connection risks.
- 4.3 Conduct a thorough on-site inspection of the water system for cross-connection prevention, including service connections, internal plumbing, and water-using equipment. Unless otherwise specifically directed by the government, the entirety of the system is to be assessed.
- 4.4 Coordinate with installation personnel to plan survey logistics, ensuring minimal disruption to daily operations. A FAFB CE Water Shop representative will be available to aid in the location of devices and access.
- 4.5 Use industry-approved methodologies to:
 - 4.5.1 Identify actual any potential cross-connections, including but not limited to:
 - 4.5.1.1 Equipment connections (e.g., boilers, irrigation systems, chemical feeders).
 - 4.5.1.2 Facility-specific risks (e.g., industrial processes, medical equipment).
 - 4.5.2 Evaluate protection provided by existing BFDs.
 - 4.5.2.1 Record device information, including model, serial number, and compliance status.
 - 4.5.2.2 Inspect all existing BFDs to:
 - 4.5.2.2.1 Verify type, installation condition, and proper placement.
 - 4.5.2.2.2 Determine if devices comply with applicable codes and standards.
 - 4.5.3 Identify deficiencies and provide corrective actions and costs.
 - 4.5.3.1 Document deficiencies related to cross-connections and BFDs, including:
 - 4.5.3.1.1 Devices non-compliant, missing, or improperly installed.
 - 4.5.3.1.2 Areas with insufficient protection or inadequate device selection.
 - 4.5.3.2 Provide detailed corrective action recommendations for all identified deficiencies.
 - 4.5.3.3 Prioritize corrective actions based on risk level and regulatory compliance requirements.
 - 4.5.3.4 Develop a detailed, itemized cost estimate and summary to address corrective actions and repairs required and recommended.
 - 4.5.3.5 Submit a comprehensive final report covered in the Deliverables section of this SOW.

5 DELIVERABLES. The contractor shall provide the following:

- 5.1 **Preliminary Survey Report**, delivered within 45 days of survey completion, detailing the initial findings and the time required to complete the survey.
- 5.2 **Final Report**, delivered within 90 days of survey completion.
 - 5.2.1 Executive summary of the survey findings including time to conduct survey and report development.
 - 5.2.2 List of identified cross-connections, deficiencies, and risk levels.
 - 5.2.3 Assessment of existing BFDs, including compliance status and risk level.
 - 5.2.4 AF Form 848 (see Appendix A for a fillable form AF Form 848) or commercial off the shelf equivalent (so long as AF Form 848 data is not missing from the equivalent and is compatible with Microsoft Office) for each building and/or potential cross-connection location, documenting results from this survey, per DAFMAN32-1067 4.3.4.8.
 - 5.2.5 Detailed corrective action recommendations, including cost and prioritization based on risk level.
 - 5.2.6 Supporting documentation in a self-explanatory, organized fashion (e.g., photographs, diagrams, and compliance references).
 - 5.2.7 Annotated maps and diagrams of the water system highlighting the locations of cross-connections and BFDs.
 - 5.2.8 Compliance assessment and matrix summarizing compliance with Air Force instructions, Washington DoH requirements, and other relevant standards.
 - 5.2.9 Detailed and itemized cost estimate summary, reports, and tables to address corrective actions and repairs identified by this survey both required and recommended, including the quotes and methods used to develop the estimates. These estimates should also include and show overhead and profit, bonds and insurance costs, and a 25% contingency.
 - 5.2.10 Editable electronic copies of the above spreadsheets (i.e. .xlsx format) and documents for future analysis, in addition to hard-copy reports, shall be delivered to the Civil Engineering representative.

6 ROLES AND RESPONSIBILITIES.

- 6.1 Government Responsibilities
 - 6.1.1 Provide point-of-contact personnel and, if available, installation water system schematics and prior survey reports.
 - 6.1.2 Ensure contractor access to all areas of the water system, including GSUs.
 - 6.1.3 Review and approve deliverables within specified timelines.
- 6.2 Contractor Responsibilities
 - 6.2.1 Plan and execute the survey in accordance with this SOW and applicable codes.

- 6.2.2 ~~All personnel~~ The team lead performing the survey shall hold the appropriate Washington State certifications in cross-connection control and backflow prevention (e.g., Washington State-certified Cross-Connection Control Specialist) and shall be the individual to sign off on all survey reports.
- 6.2.3 Ensure safety, security, and integrity of the potable water system during survey activities.
- 6.2.4 Submit required deliverables within the prescribed timeframes.

7 SAFETY REQUIREMENTS. The contractor shall adhere to:

- 7.1 Occupational Safety and Health Administration (OSHA) standards.
- 7.2 Installation-specific safety protocols.
- 7.3 Proper water quality protection measures to avoid introducing contaminants during the survey.

8 QUALITY ASSURANCE. The contractor shall:

- 8.1 Maintain compliance with all listed codes, regulations, and industry standards.
- 8.2 Ensure technical accuracy in deliverables and findings.
- 8.3 Provide feasible corrective action recommendations compliant with applicable standards.

9 SCHEDULE

- 9.1 Survey Execution: Shall be completed between 01 Jul 2026 and 30 Sep 2026
- 9.2 Preliminary Report Submission: Within 45 days of survey completion.
- 9.3 Final Report Submission: Within 90 days of survey completion.

10 GENERAL INFORMATION

- 10.1 **HOURS OF OPERATION.** Hours of operation are 0700-1630 Monday through Friday, excluding Federal Holidays and any other designated down day. Work can be done at other than the normal hours of operation with the approval of the Contracting Officer. The contractor shall coordinate in advance all hours of work and any service interruptions with Civil Engineering points of contact. Designated federal holidays and their associated down days are listed below.

January 1	New Year's Day
3rd Monday in January	Martin Luther King, Jr. Day
3rd Monday in February	President's Day
Last Monday in May	Memorial Day
June 19	Juneteenth
July 4	Independence Day
1st Monday in September	Labor Day
2nd Monday in October	Columbus Day
November 11	Veterans Day
4th Thursday in November	Thanksgiving Day
December 25	Christmas

- 10.2 **SECURITY REQUIREMENTS.** All personnel employed by the Contractor in the performance of this contract, or any representative of the Contractor entering the government installation, shall abide by all Fairchild Air Force Base security regulations.
- 10.3 **GOVERNMENT FURNISHED PROPERTY AND SERVICES.** The Government shall furnish site access, access to restroom facilities, and electrical power if inside a government-owned facility. If work is taking place exterior to a building, power is not the responsibility of the government to provide. In the event of an emergency, dial 911.

11 APPENDICES:

- 11.1 Appendix A – AF Form 848
- 11.2 Appendix B1 – Backflows Maps – Maps of Water System
- 11.3 Appendix B2 – Backflows Maps – Skils’kin Irrigation Book of Backflows (reference Appendix D for list of device locations)
- 11.4 Appendix C1 – Connections List – Annual Survey Report
- 11.5 Appendix C2 – Connections List – Builder Final 09 Report - All buildings potentially connected to the water system, includes square footage
- 11.6 Appendix D – Devices outside of ASR – Skils’kin Irrigation BFD List